

RefZVC: Refinable Zero-Shot Video Captioning by Test-Time Reinforcement Polishing

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Abstract—Recently, the zero-shot image captioning (zero-shot IC) method based on pre-trained visual language models (VLMs) and large language models (LLMs) has made significant progress. However, how to adapt it to the zero-shot video captioning (zero-shot VC) scenario (without video-text paired supervision) has not been well explored. Inspired by various recent test-time strategies (sacrificing additional test time to improve performance), we try to introduce a new paradigm of Test-time Reinforcement Polishing in zero-shot VC scenario. We take temporal dependency modeling as the starting point and propose a novel framework for Refinable Zero-shot VC, called RefZVC. RefZVC can greatly cover the long-term context of the video and continuously polish and refine the generated captions in a reward-feedback manner. We first design an Adaptive Frame Skipping module (AdaSkip) to skip redundant frames and select diverse keyframe sequences. Subsequently, we propose a Multi-granularity Reinforcement Polishing (MRP) mechanism, which iteratively polishes captions by leveraging Gaussian Kernel Cache (GKC) to capture temporal dynamics, store and reuse relevant historical context. In addition, MRP calculates rewards for generated captions at both the sentence-level and entity-level to achieve test-time polishing. With the MRP mechanism, RefZVC achieves superior zero-shot generalization performance, outperforming previous zero-shot VC methods on benchmarks such as MSVD, MSR-VTT, and VATEX.

Index Terms—Video captioning, zero-shot, reinforcement learning, test-time polish.

I. INTRODUCTION

THE goal of Video Captioning (VC) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15],

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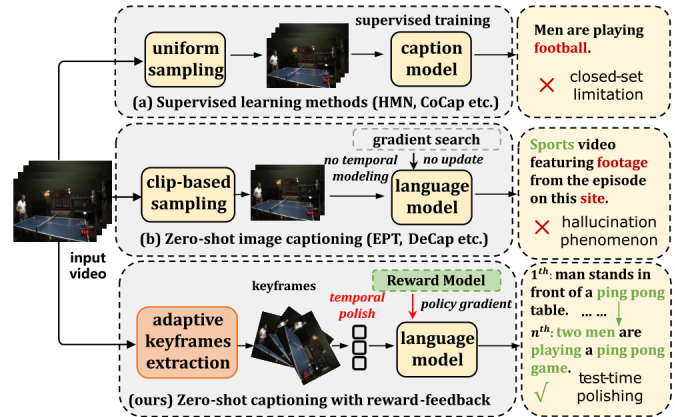


Fig. 1. Motivation comparison of different video captioning paradigms. (a) Supervised methods (HMN [2]) may suffer from closed-set limitations (misidentification as football). (b) zero-shot IC methods (DeCap [35]) without proper temporal modeling may suffer from hallucination phenomena. (c) RefZVC addresses these issues via adaptive keyframe extraction and reward-guided test-time polishing for progressive caption refinement.

[16], [17] is to automatically generate accurate and coherent descriptive text for a given video. It is a cornerstone task in the field of visual-language modeling [18], [19] and human-computer interaction [20]. Beyond caption generation, video understanding also includes recognition-oriented scenarios such as few-shot action recognition, where efficiently selecting discriminative temporal evidence is similarly important [21]. Currently, VC has made significant progress in fully supervised algorithms [4], [22], [23], [24], [25], but it has not been well explored in the zero-shot inference paradigm. The recently emerging Zero-shot Image Captioning (zero-shot IC) methods [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36] have shown us the powerful potential of using pre-trained VLMs (such as CLIP [18], etc.) and LMs (such as GPT-2 [37], OPT [38], etc.) for out-of-domain (OOD) zero-shot inference. Inspired by this, we focus on implementing Zero-shot Video Captioning (zero-shot VC) [39], [40], aiming to achieve efficient video description without using video-text paired data.

Compared with the zero-shot IC task of processing a single static image, zero-shot VC is more complex and challenging. It requires the model to flexibly handle long-term dependencies and generate logically coherent descriptions. As illustrated in Fig. 1, our motivation is to address two coupled issues in zero-shot video captioning: hallucination-prone caption generation and insufficient temporal modeling under redundant video frames. Video sequences usually contain a lot of redundant information, and the semantic changes between consecutive

frames may be very subtle. Therefore, how to filter redundancy and accurately extract key frames is crucial to the performance of zero-shot VC. As a pioneering work of zero-shot VC, EPT [39] proposed a CLIP-based frame sampling strategy to select diverse frame sequences from redundant videos, and then feed these frame embeddings into GPT-2 [37] to generate video captions. EPT used the CLIP cosine similarity between adjacent frames as the threshold criterion for frame selection, but the drawbacks of EPT are also obvious. *i)* Dense adjacent frame calculations are very time-consuming and not suitable for real-time video processing scenarios; *ii)* It is more difficult to model long-term dependencies for diverse frame sequences, thereby destroying the logical coherence and semantic consistency of generated captions (as shown in Fig. 1 (b)). Another technical route is to directly adapt the zero-shot IC method to zero-shot VC. For example, DeCap [35] abandoned the key frame selection process and instead randomly samples k frames from a given video, and used a naive temporal average pooling mechanism to obtain video-level features from frame-level features, and then decoded to generate text. Since the pooling operation may lose a lot of temporal information, it cannot well represent the temporal evolution in the video, and the quality of the generated description is not satisfactory.

Based on the above analysis, we propose the following insights: (1) adaptive keyframe selection for any given OOD video is necessary, but requires a trade-off between diversity and inference efficiency. (2) How to efficiently model the temporal dependency of the obtained keyframe sequence in the zero-shot scenario? Recently, reward feedback mechanism represented by reinforcement learning with human feedback (RLHF) [41], [42] has achieved brilliant performance in various visual-language tasks [43], [44], [45], [46], [47], [48], [49], [50]. Inspired by this, we try to introduce a new paradigm of test-time reinforcement polishing in the zero-shot video inference stage to continuously refine the generated captions to improve performance. *This process is similar to an experienced artisan constantly refining and polishing the artwork in his hands.* Based on this idea, we propose RefZVC framework, which performs multiple iterative polishing at test-time to improve performance. Specifically, at each polishing iteration, we first design an Adaptive Frame Skipping (AdaSkip) module to select diverse keyframe sequences for a given video. In order to better handle unseen video data, we convert the traditional key frame selection operation into a frame-skipping process. This allows the model to flexibly process videos containing complex events (fewer skips) and simple events (more skips). We then propose a novel Multi-granularity Reinforcement Polishing mechanism (MRP), which can iteratively polish and refine the generated captions. Since the video content itself is complex and unpredictable, when a new object appears in a video sequence, the model may use the current frame as the focus and forget the historical frames. In order to fully capture the temporal dynamics of the keyframes sequence, we maintain a Gaussian Kernel Cache (GKC) module in the zero-shot inference stage to store historical embeddings. Based on the gaussian kernel function [51], [52], GKC can adaptively select the historical information to participate in the current description to achieve a smooth transition.

Due to the lack of explicit keyframe labels and captions labels at test time, we use reinforcement learning (RL) to optimize the model. When calculating the reward signal, we use the widely recognized expert CLIP model (ViT-L/14) as the feedback source. Previous methods [47], [49], [53] usually directly calculate the CLIPScore [54] between the generated captions and the image as the reward. This type of method ignores a serious phenomenon that captions usually contain many functional words, such as “the”, “a”, “are” that are irrelevant to the visual content. This causes the reward value to ignore the objects in the captions that cause serious hallucinations (these targets usually appear as noun types). To solve this problem, we propose a Multi-Granular CLIPReward that calculates the rewards between captions and keyframe sequences at the sentence-level and entity-level at the same time. Then we optimize the model at test-time using the policy gradient algorithm. After multiple iterations of polishing, RefZVC can fully cover the temporal evolution in the video and generate more accurate descriptions under the guidance of reward feedback.

Compared with prior test-time adaptation approaches, RefZVC’s test-time reinforcement polishing framework fundamentally differs in several key aspects. While Zhao et al. [49] employ gradient-based optimization for general vision-language tasks and ZeroCap [39] focuses on image captioning with gradient search, RefZVC specifically targets video captioning using policy gradient reinforcement learning with multi-granularity rewards and GKC for temporal modeling. Unlike test-time training methods [55], [56] that update all model parameters, RefZVC selectively updates specific components (AdaSkip, projector, GKC) to maintain generalization while optimizing both frame selection policy and caption generation process through reinforcement learning. In summary, the contributions of this paper can be summarized as follows:

- We design the AdaSkip to adaptively skip frames for given video at test-time, which can achieve a better trade-off between diversity and processing efficiency.
- We propose a novel test-time reinforcement polishing framework for zero-shot video captioning. We design a novel MRP mechanism to iteratively polish captions based on multi-granular reward feedback. It can improve the zero-shot captioning performance without any labeled video-text pairs.
- Our method has excellent cross-domain zero-shot generalization ability and achieves superior performance in multiple zero-shot, in-domain and cross-domain inference settings on standard video caption datasets such as MSVD [57], MSR-VTT [1] and VATEX [6].

II. RELATED WORK

A. Supervised Video Captioning

Supervised video captioning [4], [9], [22], [58], [59], [60], [61], [62], [63], [64] is a crucial task in the field of artificial intelligence (AI) [65], [66], [67], [68], [69], [70], [71], [72], [73], [74] and video analysis [75], [76], [77], [78], [79], [80], [81]. Closely related video understanding tasks, such as group activity recognition, also require structured temporal reasoning over multiple actors and frames [82]. It aims to generate descriptive text for videos by leveraging paired video-text

datasets. This task is essential for applications such as automatic video summarization, content-based video retrieval, and multimedia understanding. Video captioning can be broadly classified into two categories: video captioning (VC) [2], [6], [8], [10], [23], [83], [84] and dense video captioning (DVC) [7], [85]. VC aims to produce a single, high-level description for an entire video, while DVC generates temporally localized captions for multiple events or scenes, providing a finer understanding of complex videos. Early approaches to this problem mainly relied on Recurrent Neural Networks (RNNs), such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) [8], [9], [59], or Transformer-based models [2], [4], [22], [23] to capture temporal dependencies within the video. These architectures aimed to model the sequential nature of video content, which is crucial for understanding how events evolve over time. However, despite their success, these early models often struggled to capture long-range dependencies and contextual information across frames, limiting the accuracy and richness of the captions generated. Subsequent advancements have focused on improving the quality of the generated captions. Techniques such as coarse-to-fine learning [9], where the model first generates rough captions and then refines them, have proven effective in enhancing the overall captioning process. This progressive refinement philosophy is also echoed by compositional action recognition studies [86], which motivates us to view iterative caption polishing from a broader video-understanding perspective. Memory augmentation [87] is another technique that incorporates external memory mechanisms to store relevant information across long video sequences, further improving the model's ability to retain important details for caption generation. Additionally, knowledge graph-based methods [83] have been introduced to enrich visual-language interactions by integrating structured world knowledge into the captioning process, enabling models to generate more contextually aware and accurate captions.

B. Zero-Shot Video/Image Captioning

Zero-shot Image captioning (zero-shot IC) addresses the challenge of generating descriptions for images or videos without access to paired training data, relying instead on Vision-Language Models (VLMs) that are pre-trained on noisy web-sourced data. Existing zero-shot image and video captioning methods can be broadly classified into two main technical approaches: training-free and text-only-training methods. The training-free approach, used by models such as ZeroCap [39] and MAGIC [30], focuses on optimizing the outputs of a pre-trained language model by performing a gradient-based search. This optimization aligns the model's generated captions with visual-textual relevance measures, such as those provided by CLIP (Contrastive Language-Image Pretraining). By doing so, these models are able to generate captions that are contextually relevant to the images or videos, without requiring additional training on specific paired datasets. On the other hand, the text-only-training approach, exemplified by models such as DeCap [35], CapDec [32], and ViECap [31], maps visual embeddings to a text-compatible space. Recent efficient multimodal and language-model inference studies further show that selecting compact visual tokens can reduce redundant context while preserving task-relevant evidence [88], which is conceptually aligned with the efficiency motivation of adaptive

frame skipping. These methods focus on enhancing zero-shot performance by utilizing only text-based training to better align visual content with textual descriptions. By mapping visual features to a space that is compatible with textual representations, these models are able to improve their ability to generate captions for previously unseen visual content. While Zero-shot Video Captioning (zero-shot VC) faces the added challenge of modeling temporal dependencies in video data, recent methods have started to address this challenge. For instance, EPT [39] partially tackles the temporal aspect by implementing a CLIP-based frame-sampling strategy, which feeds the embeddings of selected frames into a language model like GPT-2 for captioning. Similarly, the DeCap method, which originally focused on image captioning, adapts to zero-shot video captioning by applying temporal average pooling techniques. However, this method struggles to effectively represent dynamic changes across frames, leading to potential issues in capturing the full complexity of temporal interactions within the video. This limitation directly impacts the quality of the captions, as the model fails to fully account for the dynamic nature of video content.

C. Reinforcement Feedback Meets Visual-Language

Reinforcement learning with human feedback (RLHF) [42], [44] has recently gained significant attention for fine-tuning large language models (LLMs) by using reward signals derived from user pReferences. This approach has demonstrated great potential in improving model performance across a range of tasks, including visual-language tasks. Pinto et al. [43] applied reinforcement learning to vision models, enhancing object detection and segmentation capabilities. In the context of visual-language tasks, Xu et al. [48] introduced ImageReward, which uses expert preferences to optimize text-to-image generation by aligning generated visuals with textual descriptions. Similarly, CAPWAP [46] leveraged question-answer pairs as feedback to improve the relevance of generated captions. As vision-language models like CLIP continue to evolve, reinforcement learning has been increasingly applied to optimize tasks such as image captioning, classification, and retrieval [44], [47], [49], [89]. By incorporating human feedback, these models are able to more effectively capture user intent and generate more relevant, contextually accurate outputs.

III. APPROACH

A. Overview

To achieve more efficient zero-shot VC performance, remove video redundancy and improve the quality of generated captions, as shown in Fig. 2, we propose a new framework called RefZVC, which consists of two modules: AdaSkip and a language model (LM) with MRP mechanism. At each test-time polishing iteration, (i) AdaSkip is responsible for performing redundancy removal on any given video (from downstream OOD) and extracting keyframe sequences. (ii) We then deploy a pre-trained LM and feed the keyframe embeddings (extracted by the CLIP image encoder) into the LM to polish and refine the generated captions according to the MRP feedback.

B. Adaptive Frame Skipping Module

1) *Temporal Policy Modeling*: Given an input video represented as a frame sequence $\{I_t\}_{t=1}^T$, where T denotes the

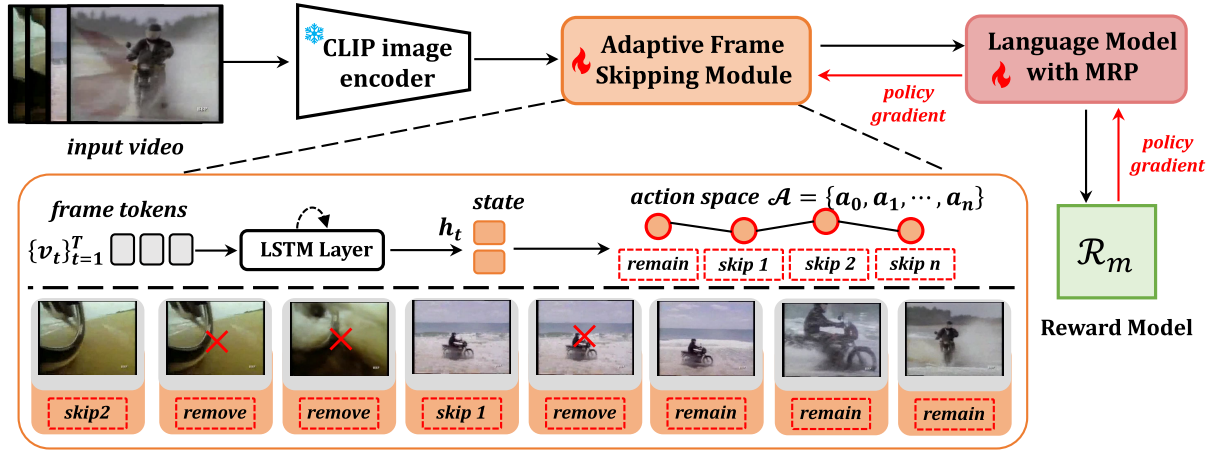


Fig. 2. The overall framework of RefZVC consists of two key modules: AdaSkip and zero-shot LM with MRP mechanism. In the zero-shot inference stage, AdaSkip can adaptively remove redundancy from the input video. Then RefZVC is updated with policy gradients based on the MRP mechanism, and the generated captions are polished and refined.

total number of frames, the goal of the AdaSkip module is to remove redundant visual content and retain only informative keyframes in a dynamic, content-aware manner. Instead of relying on fixed intervals or heuristic thresholds, AdaSkip employs a learned policy network to decide, at each timestep, whether to retain the current frame or skip ahead by a variable number of frames. To begin with, we use a frozen CLIP image encoder to extract frame-level visual embeddings: $v_t = \text{CLIP}_{\text{img}}(I_t) \in \mathbb{R}^d$ (with d as the CLIP embedding dimension). At time step $t < T$, we feed these embeddings into the policy network to generate the corresponding hidden state $\{h_t\}_{t=1}^T$ to capture the temporal dependencies between frames.

$$h_t, c_t = \text{LSTM}(v_t, h_{t-1}, c_{t-1}; \theta_{\text{LSTM}}) \quad (1)$$

where h_t is the hidden state and c_t is the cell state at time t , with θ_{LSTM} representing the LSTM learnable parameters. The LSTM output sequence $\{h_t\}$ serves as a temporal representation of each frame, capturing dependencies across the sequence.

2) *Action Space and Sampling*: Then, the policy network samples the action distribution according to the hidden state h_t to decide whether to keep the frame or skip several frames to remove redundancy. Let n be the maximum number of skipped frames, then the action space is defined as $\mathcal{A} = \{a_0, a_1, \dots, a_n\}$, where a_0 denotes that the current frame is retained, a_k (for $k = 1, 2, \dots, n$) denotes skip k frames. The action space size is therefore $n + 1$. At each time step t , the policy network computes a probability distribution $\pi(a_t = a_k | h_t; \theta)$ over possible actions based on the current state. If a_0 is selected, current frame is retained, and the policy proceeds to the next frame. If a_k (where $k \geq 1$) is selected, policy network skips k frames, and resumes action selection at time step $t + k$. This design allows the policy network to dynamically adjust the density of frame selection so that it samples densely in the information-rich parts and skips frames appropriately in the redundant parts, thereby improving computational efficiency.

C. Multi-Granularity Reinforcement Polishing

In the zero-shot inference stage, we perform L -step reinforcement learning on the given video. At each polishing iteration, AdaSkip removes redundancy from the video to obtain the keyframes embedding sequence $\{s_i\}_{i=1}^T$ (sequence

length is T). We then leverage GKC to integrate its temporal dynamics and feed it into LM to generate captions, and the proposed Multi-Granular CLIPReward is used to calculate rewards for text polishing to improve quality (as shown in Fig. 3). With continuous polishing iterations, RefZVC incorporates the temporal coherence between keyframes and the spatial alignment with the video semantics to achieve effective zero-shot captioning.

1) *Text-Only-Training LM and CLIP Feature Alignment*: We generate initial captions using a text-only pre-trained LM, following methods like CapDec [32] and ViECap [31]. Since the CLIP keyframe embeddings $\{s_i\}_{i=1}^T$ (where $s_i \in \mathbb{R}^d$) contain rich cross-modal semantic information, we can directly map these embeddings to the LM's token space using a projector $\mathcal{F}_{\text{proj}}: \mathbb{R}^d \rightarrow \mathbb{R}^m$. Here, $\mathcal{F}_{\text{proj}}$ is implemented as a transformer-based network, trained to adaptively align CLIP features with the LM's input space. At zero-shot inference time, the keyframe sequence $\{s_i\}_{i=1}^L$ can be fed into the LM to generate initial captions:

$$\hat{y}^0 = \text{LM}(\mathcal{F}_{\text{proj}}(\{s_i\}_{i=0}^T)) \quad (2)$$

where $\hat{y}^0$ denotes the initial captions, and m is the dimension of LM's token space. The LM here remains frozen during inference, while the projector $\mathcal{F}_{\text{proj}}$ is updated during the reinforcement process.

2) *Temporal Polishing via Gaussian Kernel Cache*: For video captioning tasks, it is crucial to capture temporal dependencies to accurately describe the time-evolving objects and events. To achieve this, we introduce a Gaussian Kernel Cache module $\mathcal{C}$ that stores historical information from previous polishing iterations, enabling sequential temporal polishing for current iteration. Compared to other hard correlations (such as the normalized match in cosine similarity), the gaussian kernel is more flexible and can produce a smooth and continuous similarity measure between features in different frames.

For each iteration of keyframes embedding sequence $\{s_i\}_{i=0}^T \in \mathbb{R}^{T \times d}$ and a learnable cache $\mathcal{C} \in \mathbb{R}^{1 \times d}$ (initialized to zero tensors), GKC computes a smooth similarity measure between each frame feature s_i and the cache $\mathcal{C}$ using a Gaussian-RBF kernel, defined as $\phi(s_i, \mathcal{C})$. This similarity is then used to weight the expanded cache for each frame,

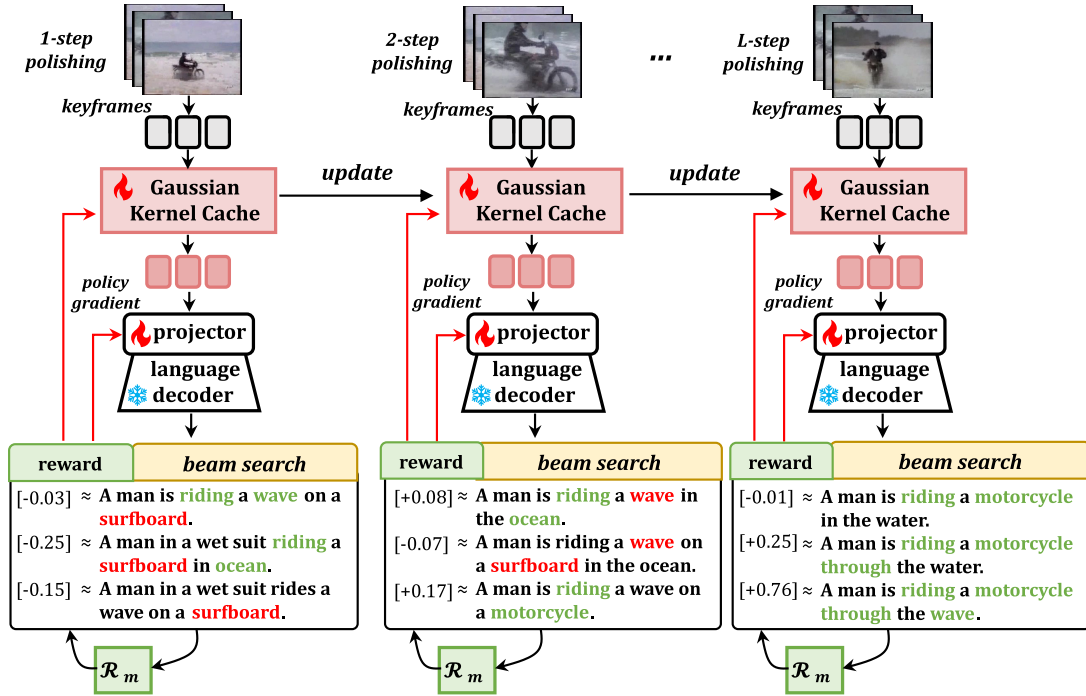


Fig. 3. The overall flow of the MRP pipeline is illustrated. At each polishing iteration, the keyframes embedded by AdaSkip is input into the GKC module for temporal polishing and then into the LM to generate captions. During generation, beam search is used for sampling to calculate the average reward. MRP uses the policy gradient algorithm to strengthen the visual consistency of content (positive reward, marked in green) in the captions and penalize the inconsistent content (negative reward, marked in red).

yielding weighted cache $\mathcal{C}_w^{(i)}$, the calculation process is as follows:

$$\phi(s_i, \mathcal{C}) = \exp\left(-\frac{\|s_i - \mathcal{C}\|^2}{2\sigma^2}\right)$$

$$\mathcal{C}_w^{(i)} = \phi(s_i, \mathcal{C}) \odot \mathcal{C} \quad (3)$$

where $\odot$ denotes element-wise multiplication, and σ is the learnable kernel bandwidth parameter. The bandwidth parameter σ governs the Gaussian kernel's sensitivity: a small σ yields peaked similarities that sharply select only the most relevant historical embeddings, whereas a large σ spreads the influence across a wider temporal window, producing smoother transitions. During reinforcement polishing σ is treated as a learnable parameter; it is initialized to 1.0 and updated by the same policy-gradient rule that adjusts the other components of Θ . Consequently, the kernel width adapts on-the-fly to the dynamics of each video, allowing the cache to flexibly emphasize either fine-grained or coarse-grained temporal context as the content demands. The weighted cache is integrated with the original frame features via a residual connection, resulting in the refined features $s'_i = s_i + \mathcal{C}_w^{(i)}$. These refined features are subsequently aggregated across the temporal dimension through average pooling, forming the polished representation $\bar{s} = \frac{1}{n} \sum_{i=1}^T s'_i$. Finally, the cache is updated via a gating mechanism that smoothly balances the previous cache with the new aggregated feature:

$$\mathcal{C}_{\text{new}} \leftarrow g \cdot \mathcal{C} + (1 - g) \cdot \bar{s} \quad (4)$$

where $g \in (0, 1)$ is a learnable scalar gate that balances prior memory and current context. This allows RefZVC to iteratively refine temporal cues across polishing rounds while preserving long-term coherence.

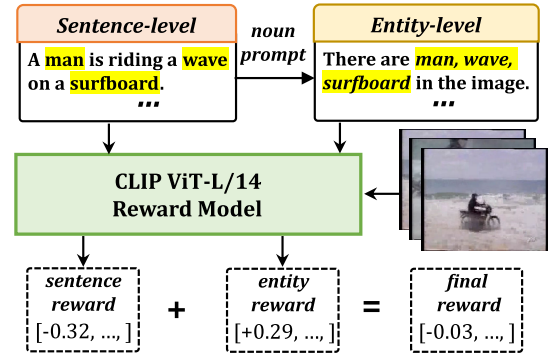


Fig. 4. Sentence-level and entity-level rewards measure the quality of the current generated text at different semantic granularity levels, and apply positive and negative rewards to the model through a baseline-adjusted strategy.

3) *Multi-Granular CLIPReward*: Due to the lack of gold video caption labels in zero-shot inference scenarios, we introduce reward feedback to calibrate and polish the generated captions. For each polishing iteration, the polished representation $\bar{s}$ is first input into the LM with beam search to generate diverse caption candidates $\{z_i^k\}_{k=1}^Q$, where Q is the beam size.

In order to align the generated captions with the visual content, previous methods [47], [49], [53] typically compute the reward signal directly as the CLIPScore (CS) [54] between the generated captions and the video frames s_i . However, these methods often suffer from the excessive appearance of function words - such as “the”, “a”, and “are”. These function words may mask the serious hallucinations exhibited by entity-level (noun) errors. To address this, we compute rewards at both the sentence-level and the entity-level (as shown in Fig. 4).

Given that CS is inherently non-negative, we introduce a baseline-adjusted reward function to penalize “hallucinations”—captions that stray from video content. This baseline adjustment subtracts the average reward across the current beam Q to allow negative feedback when captions are out of context. Specifically, for the sentence-level reward, we compute a baseline-adjusted score as:

$$\mathcal{R}_m^{\text{sent}}(\hat{z}_i, s_i) = \text{CS}(\hat{z}_i, s_i) - \frac{1}{Q} \sum_{k=1}^Q \text{CS}(\hat{z}_i^k, s_i) \quad (5)$$

In parallel, we perform noun parsing on the generated captions using the NLTK library [90] to extract noun entities, and construct a noun prompt of the form “There are $\{n_1, n_2, \dots\}$ in the image” to form an entity beam; the corresponding entity-level reward is then computed as:

$$\mathcal{R}_m^{\text{ent}}(\hat{z}_i, s_i) = \text{CS}(P(\hat{z}_i), s_i) - \frac{1}{Q} \sum_{k=1}^Q \text{CS}(P(\hat{z}_i^k), s_i) \quad (6)$$

where $P(\hat{z}_i)$ denotes the noun prompt derived from $\hat{z}_i$. The final reward signal is obtained by fusing these two granularities using a weighting parameter $\lambda = 0.8$:

$$\mathcal{R}_m(\hat{z}_i, s_i) = \lambda \mathcal{R}_m^{\text{sent}}(\hat{z}_i, s_i) + (1 - \lambda) \mathcal{R}_m^{\text{ent}}(\hat{z}_i, s_i) \quad (7)$$

To prevent overfitting and maintain the LM’s generalization capacity, the feedback mechanism restricts updates to specific parameters—the AdaSkip, the projector $\mathcal{F}_{\text{proj}}$ and the GKC $\mathcal{C}$. By isolating these components, we ensure that the LM’s broader structure remains unaffected by domain-specific information while allowing finer adjustments to support video-specific semantic coherence. Specifically, we let π_{Θ} be the generation policy controlled by parameters Θ , which includes the AdaSkip parameters Θ_{Ada} , the projector parameters $\Theta_{\mathcal{F}_{\text{proj}}}$ and cache parameters $\Theta_{\mathcal{C}}$. For each test video, we apply the REINFORCE algorithm [91] to compute the policy gradient, the final baseline-adjusted policy gradient is formulated as:

$$\nabla \Theta = \sum_{i=0}^L (\mathcal{R}_m(\hat{z}_i, s_i)) \nabla_{\Theta} \log \pi_{\Theta}(\hat{z}_i | s_i), \quad (8)$$

After L iterations of reward polishing, we can obtain high-quality video captions. Through this reward-driven iterative polishing mechanism, captions can be iteratively refined to be more semantically consistent with video frames and capture temporal contextual dynamics across frames.

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

1) *Datasets*: To adhere to the zero-shot experimentation principle, we only use text data from the MS COCO [98] training set as the pre-training corpus. For zero-shot video captioning task, we employ three widely used video captioning benchmarks: MSR-VTT [1], MSVD [57], and VATEX [6]. The MSR-VTT dataset is a standard benchmark containing 10,000 video clips, each with 20 descriptive captions. The MSVD dataset contains 1,970 videos, each with about 40 captions. Finally, the VATEX dataset provides a large-scale benchmark containing about 41,250 video clips, each with 10 English captions. Following the standard protocol, we focus on testing subsets of these datasets to evaluate the zero-shot captioning performance.

2) *Metrics*: For evaluating the generated captions’ accuracy and semantic richness, we report results using four Standard metrics: BLEU@4 (B@4) [99], CIDEr (C) [100], ROUGE (R) [101], and METEOR (M) [102].

B. Implementation Details

RefZVC employs the ViT-B/16 and ViT-B/32 versions of CLIP [18] as the basic feature backbone. For AdaSkip, we first uniformly sample 16 frames from each video and resize each frame to 224×224 . Keyframes selection is achieved through an LSTM layer with a hidden size of 512, followed by a fully connected layer and a sigmoid activation function to sample actions. For the LM, we explore two initializations: GPT-2 [37] and OPT-125M [38]. In line with methods like CapDec [32] and ViECap [31], we employ an embedding projector consisting of 8 transformer layers with a hidden size of 768 and 8 attention heads. The LM is pre-trained solely on text data from the COCO training set. We use the AdamW optimizer with a learning rate of $2e-5$, training for 15 epochs with batch size of 40 on an NVIDIA 4090 GPU. During zero-shot inference, we load the pre-trained weights of LM, and then iteratively refine the generated captions using the proposed MRP mechanism. The number of beam search samples Q is set to 5. The number of reinforcement polishing iterations L is set to 4. To guide this polishing process, we utilize a larger CLIP model, ViT-L/14, as the reward model. The policy gradient update in MRP is conducted using the AdamW optimizer with learning rate of $5e-6$, where only the parameters Θ are updated, while both the LM and CLIP parameters are frozen in all stages. For reproducibility, all reported results use the same text-only COCO pre-training corpus and the official test splits of MSR-VTT, MSVD, and VATEX. At test time, the CLIP image encoder, the reward CLIP model, and the LM are kept frozen; only AdaSkip, the projector, and GKC-related parameters are updated. The cache is initialized as zero tensors for each test video, the Gaussian bandwidth is initialized to 1.0, and the final caption is selected from beam candidates according to the multi-granular reward.

C. Different Generalization Inference Settings

We evaluate RefZVC under various inference scenarios to comprehensively assess its generalization capabilities: (1) *Zero-shot Video Captioning*, which tests the model’s ability to generate captions for videos containing unseen concepts in the target domain. In this setting, the model does not have access to any video-text pairs or visual content from the target domain during pre-training. (2) *Unpaired Video Captioning*, which includes: (a) an in-domain setting, where the model is trained and tested on the same dataset without paired video-text supervision, (b) a cross-domain setting, where training and testing occur on different datasets, allowing us to further investigate the robustness of the model’s generalization to novel video domains.

D. Setting One: Zero-Shot Video Captioning

In this section, we evaluate the performance of our proposed RefZVC against two main categories of baseline methods: i) supervised learning methods (VNS-GRU [59], SemSynAN [60], SwinBERT [22], IcoCap [92], VCRN [93], TCAT [94])

TABLE I

EXPERIMENTAL RESULTS OF ZERO-SHOT VIDEO CAPTIONING ON MSR-VTT AND MSVD. B@4, M, R, C REPRESENT BLEU@4, METEOR, ROUGE AND CIDER RESPECTIVELY. METHODS SUCH AS DeCap* [35] AND MA ET AL.* [40] USE ADDITIONAL CORPUS MEMORY AND RETRIEVAL KNOWLEDGE TO ENHANCE INFERENCE. THE BEST PERFORMANCE ON EACH METRIC IS MARKED IN BOLD

Method	Venue	MSR-VTT				MSVD			
		B@4	M	R	C	B@4	M	R	C
Paired video-text training									
VNS-GRU [59]	ResNeXt-101+GRU	45.3	29.9	63.4	53.0	66.5	42.1	79.7	121.5
SemSynAN [60]	ResNet-152+R(2+1)D+LSTM	46.4	30.4	64.7	51.9	64.4	41.9	79.5	111.5
SwinBERT [22]	VidSwin-Base + BERT _{Base}	41.9	29.9	62.1	53.8	58.2	41.3	77.5	120.6
IcoCap [92]	CLIP _{ViT-B/32} + Transformer	47.0	31.1	64.9	60.2	59.1	39.5	76.5	110.3
VCRN [93]	ResNeXt-101 + Transformer	41.5	28.1	61.2	50.2	59.1	37.4	74.6	100.8
TCAT [94]	VidSwin-Base + Transformer	48.7	31.6	65.7	63.7	66.0	43.8	81.4	147.0
Text-only-training, Zero-shot Inference									
ZeroCap [27]	CLIP _{ViT-B/32} + GPT-2 _{Base}	2.3	12.9	30.4	5.8	2.9	16.3	35.4	9.6
MAGIC [30]	CLIP _{ViT-B/32} + GPT-2 _{Base}	5.5	13.3	35.4	7.4	6.6	16.1	40.1	14.0
EPT [39]	CLIP _{ViT-B/32} + GPT-2 _{Medium}	3.0	14.6	27.7	11.3	3.0	17.8	31.4	17.4
DeCap* [35]	CLIP _{ViT-B/32} + Transformer	14.7	20.4	-	18.6	-	-	-	-
Ma et al.* [40]	CLIP _{ViT-L/14} + GPT-2 _{Medium}	3.5	17.3	28.7	22.1	5.0	24.0	36.9	38.1
Video-ChatGPT [95]	LLaVA-7B	13.8	18.2	42.5	15.1	18.5	24.3	51.2	38.2
VideoChat [96]	BLIP-2 _{ViT-G} + vicuna-7B	13.2	17.8	41.8	14.5	17.9	23.8	50.5	36.8
RETTA [97]	CLIP _{ViT-L/14} + GPT-2 _{Medium}	14.0	19.3	42.2	24.3	23.3	28.5	56.4	49.8
RefZVC (Ours)	CLIP _{ViT-B/16} + GPT-2 _{Base}	18.1	21.3	47.3	19.4	22.3	27.8	56.5	37.0
RefZVC (Ours)	CLIP _{ViT-B/32} + GPT-2 _{Base}	18.5	21.6	47.6	22.9	24.1	28.4	58.7	39.6
RefZVC (Ours)	CLIP _{ViT-B/16} + OPT-125M	19.2	21.8	47.6	25.1	26.1	30.0	59.2	49.3
RefZVC (Ours)	CLIP _{ViT-B/32} + OPT-125M	20.1	22.6	49.3	26.5	27.3	32.6	59.8	49.8

TABLE II

ZERO-SHOT VIDEO CAPTIONING RESULTS ON VATEX. B@4, M, R, C REPRESENT BLEU@4, METEOR, ROUGE AND CIDER RESPECTIVELY. † MEANS REFZVC (CLIP_{ViT-B/16} + GPT-2_{Base}), ‡ MEANS REFZVC (CLIP_{ViT-B/16} + OPT-125M). THE BEST PERFORMANCE IS MARKED IN BOLD

Methods	VATEX			
	B@4	M	R	C
Paired video-text training				
VaTeX [6]	28.4	21.7	-	45.1
HMN [2]	33.2	22.7	49.3	51.6
Zero-shot Inference				
VideoLLaMA [103]	4.3	16.3	21.8	3.8
Ma et al. [40]	2.8	14.1	23.5	23.9
DeCap [35]	13.1	15.3	-	18.7
Video-ChatGPT [95]	9.8	14.5	29.8	18.2
VideoChat [96]	9.2	14.1	29.2	17.5
RETTA [97]	11.4	16.3	32.6	23.8
RefZVC [†] (Ours)	16.6	16.0	40.5	20.3
RefZVC [‡] (Ours)	17.9	17.1	41.6	24.8

and *ii*) zero-shot methods (ZeroCap [27], MAGIC [30], EPT [39], VideoLLaMA [103], Ma et al. [40], DeCap [35], Video-ChatGPT-7B [95], VideoChat-7B [96], and RETTA [97]). The results are shown in Table I and Table II.

Since RefZVC is trained exclusively on COCO's text data, without access to any video-text pairs, it still has a certain gap in various metrics compared to supervised methods. Within zero-shot baselines, RefZVC demonstrates significant improvements over training-free methods such as ZeroCap, MAGIC, and EPT, as reflected in higher scores across most metrics, including B@4, C and R. Among them, the performance of RefZVC (CLIP_{ViT-B/32} + OPT-125M) is superior to other backbone versions of the model, and it has achieved SOTA on the CIDER indicator of all datasets. Training-free methods (ZeroCap, MAGIC, EPT) often suffer from the limitations of large language models, including hallucinations and poor temporal understanding, leading to fragmented or inconsistent descriptions. Although the text-only-training method represented by DeCap can alleviate the hallucination phenomenon, it only uses simple temporal average pooling to process sequence features, which loses the temporal dependency. This leads to poor performance in metrics such as M and C that better reflect the semantic integrity and coherence of generated captions. Our RefZVC also belongs to the text-only-training route. Relying on AdaSkip and MRP mechanisms, it can filter temporal redundancy for efficient temporal modeling. We also compared RefZVC with recent state-of-the-art zero-shot methods: Video-ChatGPT [96] and VideoChat [96] are multimodal large-scale language models that leverage prompts for video understanding. RETTA [97] employs retrieval-enhanced test-time adaptation to improve zero-shot video caption generation performance. RefZVC outperforms these baseline methods on all datasets. Overall, RefZVC has

TABLE III

IN-DOMAIN VIDEO CAPTIONING RESULTS ON MSR-VTT AND VATEX. ALL METHODS IN THIS TABLE USE TEXT-ONLY TRAINING. [†] MEANS REFZVC (CLIP_{VIT-B/16} + GPT-2_{BASE}), [‡] MEANS REFZVC (CLIP_{VIT-B/16} + OPT-125M). THE BEST PERFORMANCE ON EACH METRIC IS MARKED IN BOLD

Methods	MSR-VTT				VATEX			
	B@4	M	R	C	B@4	M	R	C
CLIPRe [35]	10.2	18.8	-	19.9	11.1	17.0	-	27.1
DeCap [35]	23.1	23.6	-	34.8	21.3	20.7	-	43.1
RefZVC [†] (Ours)	19.6	22.6	48.6	33.9	17.9	19.8	42.4	36.9
RefZVC [‡] (Ours)	26.8	23.4	51.2	35.7	26.8	21.2	46.6	44.0

TABLE IV

CROSS-DOMAIN VIDEO CAPTIONING RESULTS ON MSR-VTT AND MSVD. CLIPRe* AND DeCap* ARE OUR REIMPLEMENTATION. [†] MEANS REFZVC (CLIP_{VIT-B/16} + GPT-2_{BASE}), [‡] MEANS REFZVC (CLIP_{VIT-B/16} + OPT-125M). THE BEST PERFORMANCE IS MARKED IN BOLD

Methods	MSRVTT → MSVD				MSVD → MSRVTT			
	B@4	M	R	C	B@4	M	R	C
Paired video-text training								
CoCap [4]	39.5	34.3	68.7	71.2	28.7	22.8	54.9	29.7
SwinBERT [22]	46.4	37.2	71.6	81.7	31.1	24.7	56.3	34.4
Text-only-training, zero-shot inference								
CLIPRe* [35]	8.7	14.2	25.5	27.9	9.4	12.1	20.4	15.3
DeCap* [35]	19.4	21.8	40.5	39.1	18.3	17.3	39.4	19.3
RefZVC [†] (Ours)	20.0	26.1	54.0	37.4	22.6	21.6	51.5	24.0
RefZVC [‡] (Ours)	25.1	27.8	56.5	45.2	22.9	21.2	51.8	24.3

established a new standard for zero-shot video captioning by effectively balancing temporal dependency modeling and a powerful reward feedback mechanism, significantly improving the quality and coherence of generated captions.

Statistical significance is rigorously validated through paired t-tests. On MSR-VTT, RefZVC surpasses DeCap with a CIDEr improvement of $p < 0.01$ and a METEOR gain of $p < 0.05$; on MSVD, the CIDEr advantage reaches $p < 0.001$ and BLEU@4 $p < 0.01$. Compared with the fully-supervised SwinBERT, RefZVC still achieves statistically significant in-domain boosts in CIDEr and ROUGE ($p < 0.05$). When transferred to the VATEX domain, RefZVC retains significant cross-domain superiority ($p < 0.05$), confirming the robustness of the proposed approach.

E. Setting Two: Unpaired Video Captioning

In this setting, we assess the performance of RefZVC under both in-domain and cross-domain scenarios. The experimental results are shown in Table III and Table IV, respectively.

1) *In-Domain Captioning*: In this study, we only use the training text data of MSR-VTT or VATEX for pre-training. We compare RefZVC with two unpaired baselines, CLIPRe [35] and DeCap [35]. They both incorporate additional memory modules to store the pre-trained text corpus and perform retrieval enhancement at inference time. But they are deficient in temporal modeling and only obtain video-level features

through naive average pooling. RefZVC has obvious advantages in metrics such as B@4, M and C, demonstrating the effectiveness of GKC in temporal modeling.

2) *Cross-Domain Captioning*: For cross-domain scenarios, RefZVC is benchmarked against two state-of-the-art supervised methods, SwinBERT [22] and CoCap [4]. While these supervised models perform well in their training domains, their performance drops notably in cross-domain settings due to the domain-specific dependencies. In contrast, RefZVC achieves comparable performance to supervised methods in some metrics (e.g., up to 21.6 METEOR and 51.8 ROUGE for MSVD→MSRVTT) without paired video-text supervision. In addition, we also reimplement CLIPRe [35] and DeCap [35] in the same cross-domain setting, and the results show that RefZVC significantly outperforms these two text-only-training baselines. This suggests that RefZVC can show better generalization ability in the cross-domain setting.

F. Qualitative Analysis

Fig. 5 shows the intermediate process of RefZVC training on the MSR-VTT dataset. In the first example, the video content shows a baseball game. First, the AdaSkip module selects a series of key frames from the input frame sequence, and these selected key frames are marked with red boxes. The content in the dotted box shows the polishing and reward results of RefZVC in the process of generating captions (we omitted some intermediate steps). In the intermediate polishing results, although RefZVC mistakenly identified the flying baseball as a “kite” at a certain moment, the MRP mechanism quickly corrected this. Through the feedback of negative rewards (−0.0176), RefZVC punished and adjusted this error to ensure the accuracy of the caption content. In addition, RefZVC can still retain the historical information of “throwing” and “baseball” in the previous frame during the processing process, which reflects its strong temporal modeling capabilities and sensitivity to temporal context. In this way, RefZVC can effectively capture the temporal dependencies in the video and generate more coherent and accurate captions.

In the third example (children decorating a cake), the intermediate outputs further reveal the motivation of our polishing strategy. At early iterations, the model produces partially mismatched entities (e.g., “pizza”), and the reward sequence contains negative values (e.g., + 0.0118, −0.0040, −0.0240), indicating semantic inconsistency with video content. As polishing proceeds, the generated keywords/caption fragments gradually shift toward semantically correct concepts (e.g., “kids”, “decorating”, “cake”), and the reward trend becomes predominantly positive (e.g., −0.0193, + 0.0278, + 0.0013, then + 0.0310, + 0.0020, + 0.0050). This example demonstrates that reward-guided test-time polishing can effectively suppress hallucinated entities and improve temporal-semantic consistency.

G. Human and LLM Evaluation

To complement automatic metrics, we further conduct preference-based evaluation from both human judges and an external LLM judge. We randomly sample 100 videos from the MSR-VTT test set and build pairwise comparison sets between RefZVC and two representative zero-shot baselines, EPT [39] and DeCap [35]. For each video, evaluators are shown the

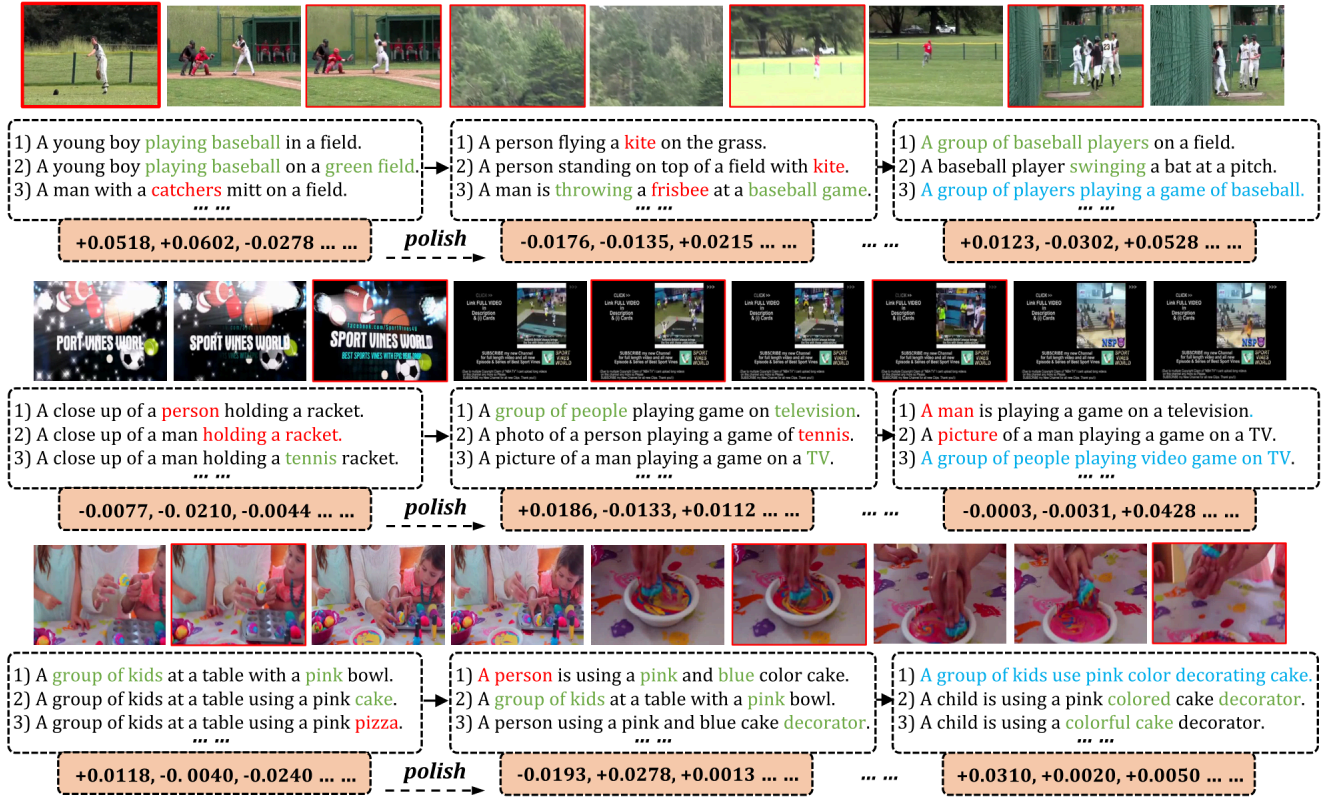


Fig. 5. Visualization of RefZVC’s intermediate process. The keyframes selected by AdaSkip are outlined in red. Only 3 candidates are shown here (correct description is marked green and incorrect description is marked red). The final generated caption is marked in blue.

TABLE V

HUMAN AND LLM EVALUATION SCORES. THE PERCENTAGES INDICATE HOW OFTEN THE CAPTION GENERATED BY REFZVC IS PREFERRED OVER THE BASELINE METHOD

Method	Human	GLM-4 [104]
EPT [39]	48.5%	47.5%
RefZVC (Ours)	51.5%	52.5%
DeCap [35]	46.0%	45.5%
RefZVC (Ours)	54.0%	54.5%

same video clip and two candidate captions, and are asked to answer a single question: “Which caption is more consistent with the video content?” The order of the two candidate captions is randomized to reduce position bias, and method identities are hidden during evaluation. For human evaluation, 5 volunteers independently complete the pairwise judgments, and we report the final preference rate after aggregation. For LLM evaluation, we use GLM-4 [104] with the same comparison protocol and prompt template, so that the two settings remain directly comparable. As reported in Table V, RefZVC is consistently preferred over both baselines by human evaluators and GLM-4, indicating stronger semantic consistency and better overall caption quality.

H. Ablation Studies

In this section, we perform a series of ablation studies to evaluate the contributions of various components in the proposed RefZVC framework. The experiments are

conducted on the MSR-VTT dataset for zero-shot video captioning, and we evaluate two versions of RefZVC: RefZVC⁺ (CLIP_{ViT-B/16} + GPT-2_{Base}) and RefZVC[‡] (CLIP_{ViT-B/16} + OPT-125M), with and without different components, including the Multi-Granular Reinforcement Polishing (MRP) mechanism and the Gaussian Kernel Cache (GKC) module.

1) *Impact of Different CLIP or LM Backbones:* We begin by evaluating the influence of the backbone models, namely CLIP_{ViT-B/16} with GPT-2_{Base} (RefZVC⁺) and CLIP_{ViT-B/16} with OPT-125M (RefZVC[‡]), on the overall performance. As shown in Table VI, both backbone combinations achieve significant improvements over their respective baselines. For RefZVC⁺, the baseline model (CLIP_{ViT-B/16} + GPT-2_{Base}) performs poorly with a BLEU@4 score of 16.8, METEOR of 20.1, ROUGE of 45.3, and CIDEr of 17.5. In contrast, RefZVC⁺ improves across all metrics, with a BLEU@4 of 18.1, METEOR of 21.3, ROUGE of 47.3, and CIDEr of 19.4, demonstrating the effectiveness of the reinforcement polishing mechanism. Similarly, RefZVC[‡] also shows a similar trend. This indicates that our RefZVC can significantly improve the performance regardless of the backbone model adopted.

2) *Ablation Studies on MRP Mechanism:* Next, we analyze the effect of the MRP mechanism by performing ablation studies that evaluate its impact on both sentence-level and entity-level rewards. These two granular rewards are crucial for refining the generated captions, as shown by the results in Table VI. For RefZVC⁺, when we remove the MRP mechanism and only use entity-level rewards, performance improves slightly (BLEU@4 = 17.8, METEOR = 20.8) compared to

TABLE VI

ABLATION EXPERIMENTS ON VARIOUS COMPONENTS AND BACKBONES IN MULTI-GRANULARITY REINFORCEMENT POLISHING (MRP) MECHANISM AND GAUSSIAN KERNEL CACHE (GKC) ON MSR-VTT AND MSVD. WE PERFORMED ABLATION ON TWO VERSIONS OF REFZVC: REFZVC[†] (CLIP_{VIT-B/16} + GPT-2_{Base}) AND REFZVC[‡] (CLIP_{VIT-B/16} + OPT-125M). B@4, M, R, C REPRESENT BLEU@4, METEOR, ROUGE AND CIDER RESPECTIVELY

Method	MSR-VTT				MSVD			
	B@4	M	R	C	B@4	M	R	C
RefZVC[†] means RefZVC (CLIP_{VIT-B/16} + GPT-2_{Base})								
Baseline [†] (CLIP _{VIT-B/16} + GPT-2 _{Base})	16.8	20.1	45.3	17.5	20.5	25.6	55.1	35.8
RefZVC [†] w/o MRP (only entity-reward)	17.8	20.8	46.8	18.8	21.9	26.9	56.0	36.3
RefZVC [†] w/o MRP (only sentence-reward)	18.0	20.9	47.0	19.0	22.1	27.4	56.2	36.8
RefZVC [†] w/o GKC (Avgpool)	17.2	20.5	46.5	18.2	21.4	26.2	55.9	36.0
RefZVC [†] w/o GKC (LSTM)	16.8	20.6	46.0	18.5	20.8	26.0	56.1	35.6
RefZVC [†] w/o GKC (Self-Attention)	17.2	19.9	46.1	18.4	20.5	25.8	55.1	35.8
RefZVC [†] (Ours)	18.1	21.3	47.3	19.4	22.3	27.8	56.5	37.0
RefZVC[‡] means RefZVC (CLIP_{VIT-B/16} + OPT-125M)								
Baseline [‡] (CLIP _{VIT-B/16} + OPT-125M)	18.4	20.6	46.2	20.9	22.5	28.7	57.1	46.8
RefZVC [‡] w/o MRP (only entity-reward)	18.7	20.5	46.9	23.9	25.0	28.4	57.0	48.2
RefZVC [‡] w/o MRP (only sentence-reward)	19.0	21.0	47.1	24.1	25.4	29.5	58.6	48.5
RefZVC [‡] w/o GKC (Avgpool)	18.6	20.8	47.0	22.5	25.5	29.5	58.8	48.0
RefZVC [‡] w/o GKC (LSTM)	18.5	20.1	45.9	22.4	25.4	28.9	58.2	47.9
RefZVC [‡] w/o GKC (Self-Attention)	18.0	20.3	46.8	21.3	24.8	29.0	58.1	47.5
RefZVC [‡] (Ours)	19.2	21.8	47.6	25.1	26.1	30.0	59.2	49.3

the baseline. However, the performance is still far below that of the full model with both rewards. A similar trend is observed when only sentence-level rewards are applied, with BLEU@4 improving to 18.0 and METEOR to 20.9. However, the best performance is achieved when both rewards are utilized together.

3) *Ablation Studies on GKC Module*: Subsequently, we evaluate the contribution of the GKC module in temporal modeling. To explore this, we replace GKC with other naive temporal modeling mechanisms, including an average pooling layer, two LSTM layers, and two self-attention layers. The performance of different configurations is summarized in Table VI. For RefZVC[†], replacing GKC with average pooling leads to a slight performance drop, with BLEU@4 decreasing from 18.1 to 17.2, METEOR from 21.3 to 20.5, and CIDEr from 19.4 to 18.2. Similarly, replacing GKC with LSTM or self-attention layers also results in performance degradation, with BLEU@4 dropping to 16.8 and 17.2, respectively, compared to 18.1 for the full model. These results highlight that GKC's unique approach to capturing temporal dependencies and incorporating a memory cache significantly improves the temporal coherence of generated captions. A similar pattern is observed for RefZVC[‡]. The performance drop is less severe than for RefZVC[†], but it still shows the importance of the GKC module.

4) *Impact of AdaSkip Module and Inference Speed Analysis*: In this section, we verify the necessity of the AdaSkip module on both the quality of generated captions and the inference speed. To this end, we compare AdaSkip with two other sampling methods: random frame sampling and the CLIP-based sampling method proposed by EPT [39]. We evaluate these methods on the MSR-VTT dataset using the RefZVC[‡] (CLIP_{VIT-B/16} + OPT-125M) version and report both

TABLE VII

ABLATION STUDY AND INFERENCE SPEED ANALYSIS OF ADASkip ON THE MSR-VTT DATASET. WE REPORT THE AVERAGE TIME TO PROCESS A VIDEO ON A 4090 GPU. THE REFZVC[‡] (CLIP_{VIT-B/16} + OPT-125M) VERSION IS USED

Methods	MSR-VTT				
	B@4	M	R	C	Time
EPT [39]	3.0	14.6	27.7	11.3	10200ms
DeCap [35]	14.7	20.4	-	18.6	2080ms
RefZVC [‡] (random)	18.5	20.9	46.8	22.9	2980ms
RefZVC [‡] (CLIP-based)	18.6	21.4	47.1	23.5	8420ms
RefZVC [‡] (AdaSkip)	19.2	21.8	47.6	25.1	3010ms

captioning performance and processing time (We randomly selected 100 videos in MSR-VTT and calculated the average inference time on a single NVIDIA 4090 GPU).

The results of our ablation study are summarized in Table VII. The baseline methods: random frame sampling and the CLIP-based method. Random frame sampling, which selects 16 frames at random from a video, achieves a BLEU@4 score of 18.5, METEOR of 20.9, ROUGE of 46.8, and CIDEr of 22.9. While these results are reasonable, they are significantly lower than those achieved by AdaSkip. The CLIP-based sampling method, which selects frames based on the proximity of their features to an anchor frame using a cosine similarity threshold $\lambda_{\cos} = 0.9$, improves the results slightly. While this method offers a more selective approach to frame sampling, it still falls short of AdaSkip's performance.

In terms of processing time, random frame sampling takes 2980 ms per video, while the CLIP-based method, requiring additional computation for cosine similarity calculations,

TABLE VIII

ABLATION STUDY ON THE WEIGHTING PARAMETER λ IN MULTI-GRANULAR CLIPREWARD. THE EXPERIMENTS ARE CONDUCTED ON MSR-VTT AND MSVD DATASETS USING REFZVC $\ddagger$ (CLIP_{VIT-B/16} + OPT-125M)

λ	MSR-VTT				MSVD			
	B@4	M	R	C	B@4	M	R	C
0.5	18.3	21.2	46.8	23.2	25.4	29.2	58.5	47.8
0.6	18.6	21.4	47.1	23.9	25.7	29.5	58.8	48.3
0.7	18.9	21.6	47.8	24.5	25.9	29.8	59.3	48.8
0.8	19.2	21.8	47.6	25.1	26.1	30.0	59.2	49.3
0.9	19.0	21.7	47.5	24.8	26.0	30.2	59.1	49.1
1.0	18.8	21.5	47.3	24.5	25.8	29.7	58.9	48.9

TABLE IX

COMPUTATIONAL OVERHEAD ANALYSIS ON MSR-VTT DATASET. WE REPORT PEAK GPU MEMORY USAGE, GFLOPS, AND PERFORMANCE-PER-GFLOPS RATIO (CIDEr/GFLOPS). AVERAGE VIDEO LENGTH IS 10 SECONDS AT 30 FPS

Methods	Computational Metrics		
	Memory (GB)	GFLOPS	CIDEr/GFLOPS
DeCap [35]	3.5	3.0	6.2
EPT [39]	4.5	1.9	5.8
RefZVC $\dagger$ (CLIP _{VIT-B/16} + GPT-2 _{Base})	4.2	3.1	6.3
RefZVC $\ddagger$ (CLIP _{VIT-B/16} + OPT-125M)	3.8	2.3	10.9

takes 8420 ms. Despite its dynamic frame selection, AdaSkip maintains a competitive processing time of 3010 ms, nearly identical to random sampling. This demonstrates that AdaSkip significantly improves captioning quality without increasing computational cost, making it a more efficient choice compared to the CLIP-based approach.

5) *Ablation Study on Weighting Parameter λ* : We also conduct an ablation study on the weighting parameter λ in the Multi-Granular CLIPReward to understand its impact on caption quality. The weighting parameter λ controls the balance between sentence-level and entity-level rewards. As shown in Table VIII, we evaluate different λ values ranging from 0.5 to 1.0 on both MSR-VTT and MSVD datasets using RefZVC $\ddagger$ (CLIP_{VIT-B/16} + OPT-125M). The results show that $\lambda = 0.8$ achieves the best overall performance across most metrics (BLEU@4: 19.2, CIDEr: 25.1 on MSR-VTT; BLEU@4: 26.1, CIDEr: 49.3 on MSVD). When λ is too small (e.g., 0.5), the model focuses more on entity-level rewards, which helps reduce hallucinations but may compromise sentence fluency. When λ is too large (e.g., 1.0), the model focuses more on sentence-level rewards, which improves fluency but may increase entity hallucinations. The optimal value $\lambda = 0.8$ provides a good balance between maintaining caption fluency and reducing entity hallucinations, demonstrating the effectiveness of the multi-granularity reward design.

6) *Computational Overhead Analysis*: We analyze the computational overhead of RefZVC in terms of GPU memory usage and GFLOPS (as shown in Table IX). RefZVC $\dagger$ and RefZVC $\ddagger$ consume 4.2 GB and 3.8 GB GPU memory, respectively, mainly for the Gaussian Kernel Cache and beam candidates, while the full model costs 2.3 GFLOPS per video and delivers the best CIDEr/GFLOPS efficiency, confirming that the modest overhead is well paid off by the performance

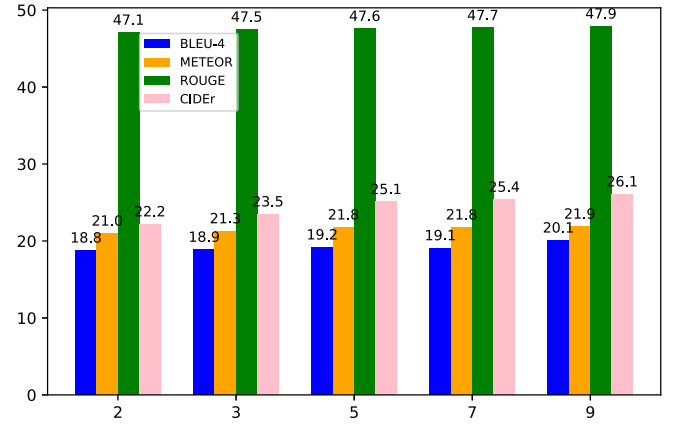


Fig. 6. The impact of different beam search size (Q) on the MSR-VTT dataset. The horizontal axis represents the range of beam search sizes, and the vertical axis represents the scores of B@4, M, R, and C. The RefZVC $\ddagger$ (CLIP_{VIT-B/16} + OPT-125M) version is used.

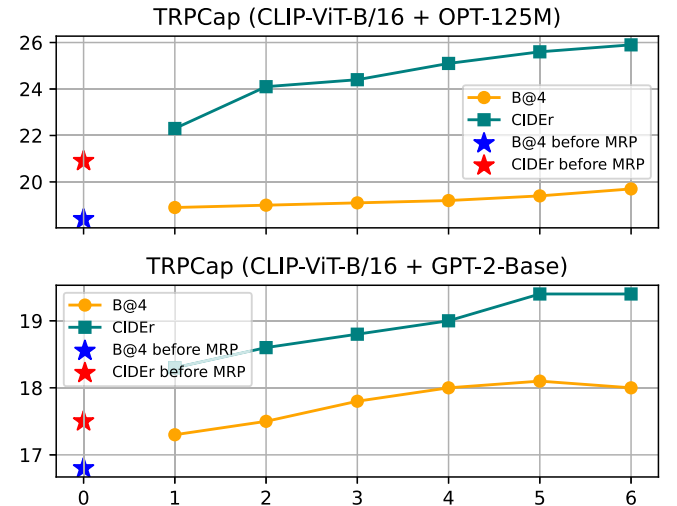


Fig. 7. The impact of different polishing iterations (L) on the MSR-VTT dataset. The horizontal axis represents the range of polishing iterations, and the vertical axis represents the scores of B@4 and C.

gain. The additional cost introduced by multiple polishing iterations is mainly from repeated reward evaluation and policy-gradient updates, and therefore grows approximately linearly with L . In practice, the default setting $L = 4$ provides a favorable balance: it substantially improves caption quality while keeping the average per-video processing time close to random frame sampling, as reported in Table VII.

7) *Impact of Beam Size Q and Polish Iterations L* : Finally, we investigate the effect of the beam search size Q and the number of polish iterations L on the performance of our model in video captioning, specifically focusing on the quality of the generated captions. We perform experiments on the MSR-VTT dataset and report the results in Fig. 6 and Fig. 7.

As shown in the results, both the beam size Q and the number of polish iterations L contribute positively to the captioning quality. As Q and L increase, the BLEU@4 and CIDEr scores show a consistent improvement. This is expected, as larger beam sizes allow for more diverse and accurate

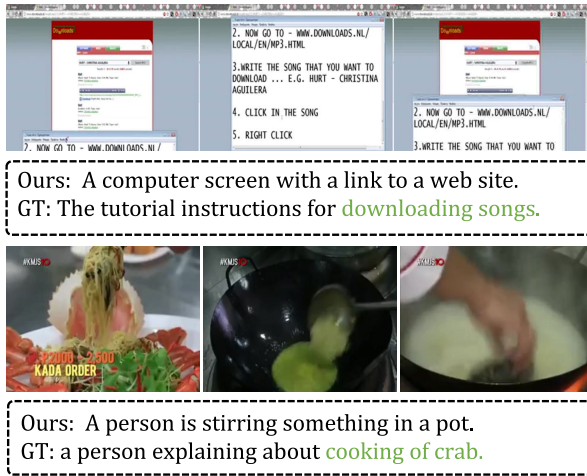


Fig. 8. Typical failure cases of RefZVC.

candidate captions during the decoding process, while multiple polish iterations refine the generated captions, leading to better fluency and content alignment with the video. For example, in the case of the RefZVC (CLIP_{VIT-B/16} + OPT-125M) model, we observe that the BLEU@4 score increases from 18.9 (with $L = 1$) to 19.7 (with $L = 6$), and the CIDEr score increases from 22.3 to 25.9. This trend illustrates that the additional polish iterations help the model refine its predictions, enhancing both syntactic quality and semantic relevance. However, it is important to note that while increasing Q and L does improve the captioning performance, the gains are not infinite. For very large values of Q and L , the improvements become marginal, and computational overhead grows significantly. In practice, there is a trade-off between performance and computational cost. Therefore, while larger values of Q and L can be beneficial, the choice of optimal values depends on balancing quality and inference speed. We recommend the following configurations for different deployment scenarios. For applications requiring a balance between image quality and inference speed, we suggest setting the beam search size $Q = 5$ and polishing iterations $L = 4$ to the optimal balance. For scenarios where image quality is critical and computational resources are plentiful, we recommend using $Q = 8$ and $L = 5$. Conversely, for real-time applications or resource-constrained environments, a lightweight configuration of $Q = 3$ and $L = 2$ can be used.

V. ERROR ANALYSIS

In addition to successful examples, we further analyze typical failure cases to clarify the current limitations of RefZVC. As shown in Fig. 8, two challenging scenarios remain difficult: (a) Lack of OCR capability: when videos contain dense textual content (e.g., tutorial screens), RefZVC can only provide coarse descriptions and may miss key text semantics because no dedicated OCR module is used. (b) Low-resolution or motion-blurred videos: when visual evidence is weak, RefZVC may capture global actions but fail to identify fine-grained entities (e.g., missing the specific ingredient in a cooking video). In addition, RefZVC relies on CLIP-based reward feedback, so its polishing process may inherit the recognition bias of the reward model when encountering rare objects,

domain-specific terminology, or highly specialized scenes. For very long videos, repeated polishing also increases latency, and temporal relations spanning multiple distant events may still be under-modeled by the current cache design. These observations indicate that improving text reading ability and robustness to low-quality visual signals is important for future work.

VI. APPLICATIONS AND FUTURE DIRECTIONS

The zero-shot capabilities of RefZVC enable a wide range of practical applications, ranging from enhancing content-based video retrieval and searchability in unlabeled databases to providing real-time descriptive accessibility for visually impaired users. Furthermore, its ability to generate high-quality descriptions without task-specific training facilitates efficient automated content moderation and large-scale video analytics. Beyond the scope of VLM-based captioning, the central ideas of adaptive evidence selection and test-time reward polishing may also benefit broader video and multimodal understanding scenarios, such as dense audio-visual event detection [105]. Together with related action recognition and efficient visual-token selection studies discussed above, these tasks share a common need to select informative temporal or multimodal evidence under limited supervision, which is consistent with the motivation of AdaSkip and MRP. Despite these advantages, several avenues for future research remain; specifically, addressing the computational overhead of processing long-form content through hierarchical scene segmentation and optimizing the latency of the iterative polishing mechanism. Another promising direction is to integrate RefZVC with larger video-language foundation models and stronger multimodal reward models, so that the polishing framework can exploit richer temporal grounding while preserving zero-shot generalization. Future work will focus on parallelizing frame processing and streamlining reward computation to achieve true real-time performance across diverse video lengths.

VII. CONCLUSION

In this paper, we comprehensively tackle the challenges of zero-shot video caption generation using pre-trained VLMs and LMs. We present RefZVC to generate coherent video captions without paired video-text data. RefZVC uses the Adaptive Frame Skipping (AdaSkip) module to filter redundant frames, which not only reduces computational load but also retains key visual information. Then, the Multi-Grained Reinforcement (MRP) mechanism is used to refine subtitles through iterative feedback at multiple levels of granularity, including sentence and entity rewards. In this process, the Gaussian Kernel Cache (GKC) is used to model temporal dynamics, thereby enhancing the continuity and coherence of the generated subtitles. Extensive experiments show that our RefZVC achieves competitive performance in zero-shot video captioning, which is a novel and valuable attempt in the field of zero-shot captioning.

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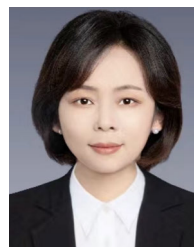


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